

EDUCATION

- **University of Texas at Austin** Austin, TX
Master of Science in Electrical and Computer Engineering; GPA: 3.86 Aug. 2024 – Present
- **Birla Institute of Technology and Science** Pilani, India
B.E in Electrical and Electronics; M.Sc. in Chemistry; GPA: 8.86/10 (top 10% of the class) Aug. 2017 - July. 2023

TECHNICAL SKILLS

- **Areas of Interest:** Robotics, Reinforcement Learning, TAMP, LLM-assisted robotics
- Proficient in::** Python, PyTorch, Keras, Git, L^AT_EX, C++, Docker, ROS/ROS2
- Robotic Simulators:** [IsaacLab](#), [Gazebo](#), [Pybullet](#), [Coppeliassim](#), [AI2THOR](#), [AIHabitat](#), [PDDLStream](#)

RESEARCH EXPERIENCE

- **University of Texas at Austin** [Learning Agents Research Group](#)
Graduate Research Assistant June 2024 - Present
 - **LLM assisted Task and Motion Planning:** [🔗](#)
 - * Developed CoCo-TAMP, a hierarchical state estimation robot framework that integrates Large Language Models with Task and Motion Planning to solve long horizon robot tasks in a partially observable environment.
 - * Engineered a belief-shaping mechanism using LLM-derived common-sense priors, reducing the POMDP search space and achieving a 67% reduction in execution time for long-horizon tasks on the Toyota HSRB.
 - * Work accepted at ICRA 2026
 - **Anticipatory Reinforcement Learning for Improving Continual Planning in homes.:**
 - * Devised a task-conditioned Markov Decision Process that anticipates future tasks and optimizes anticipated tasks using Deep Reinforcement Learning.
 - **RoboCup@Home** [🔗](#) :
 - * Team member of the UT Austin Villa@Home team for 2025 and 2026 RoboCup@Home competition.
 - * Built a unified system for general-purpose service robot that can work in real world conditions under different lighting and in crowded environments.
 - * Our team finished 2nd place in the Domestic Standard Platform League (DSPL) in 2025.
 - * Developed working mobile manipulation system with sim-to-real transfer for different robot embodiments: Toyota HSRB, Booster T1 Humanoid, and custom built robots.
- **IIIT Hyderabad** [Robotics Research Centre](#), Hyderabad
Research Engineer Oct 2022 - April 2024
Direct Supervisor: [Prof. Madhava Krishna](#)
 - **CLIPGraphs** [🔗](#):
 - * Integrated graph networks with multimodal input data to develop commonsense reasoning. Objects are mapped to their correct rooms for the rearrangement task in Embodied AI.
 - * Trained the graph network to generate an embedding space that efficiently represents the similarity between graph nodes.
 - * Presented the work at IEEE RO-MAN 2023, and EEML Summer School 2023.
 - **Anticipate & Act** [🔗](#):
 - * This project aims to create an intelligent household agent that can learn the pattern of task execution.
 - * Used and compared different Large Language Models for extracting user behavior and preferences.
 - * Created representations of tasks and actions in Planning Domain Definition Language (PDDL).
 - * Paper presented at IEEE ICRA 2024.
- **University of Luxembourg** [Theoretical Chemical Physics](#), Luxembourg
Research Assistant June 2021 - Oct 2022
Supervisor: [Prof. Alexandre Tkatchenko](#)
 - **Deep learning for quantum chemistry:**
 - * Developed novel molecular descriptors to mathematically encode molecules. Existing geometric descriptors are combined with electronic properties to generate appropriate molecular representation.
 - * Designed a representation model using Autoencoders to reduce the dimensionality of high-dimensional geometric descriptors like SLATM, and Bag-of-bonds, and a prediction model to predict molecular properties.

PUBLICATIONS

- **Large-Language-Model-Guided State Estimation for Partially Observable Task and Motion Planning**  : Y. Kim, **Raghav Arora**, R. Martín-Martín, P. Stone, B. Abbatematteo, Y. Sung
In: ICRA 2026 ()
- **Harnessing Discrete Representations for Bayesian Inference of Gene Regulatory Networks with GFlowNets**: N. Pande, **Raghav Arora**, D. Liu
In: Women in Machine Learning Workshop at NeurIPS 2024 ()
- **Anticipate & Act: Integrating LLMs and Classical Planning for Efficient Task Execution in Household Environments**  : **Raghav Arora**, S. Singh, K. Swaminathan, A. Datta, S. Banerjee, B. Bhowmick, K. Jatavallabhula, M. Sridharan, and M. Krishna
In: IEEE International Conference on Robotics and Automation (ICRA) 2024 () . Yokohama, Japan.
- **Anticipate & Collab: Data-driven Task Anticipation and Knowledge-driven Planning for Huma-robot Collaboration**  : S. Singh*, K. Swaminathan*, **Raghav Arora***, R. Singh, A. Datta, S. Banerjee, D. Das, M. Sridharan, and M. Krishna
In: Cooking Robotics Workshop at ICRA 2024 () . Yokohama, Japan.
- **CLIPGraphs: Multimodal Graph Networks to Infer Object-Room Affinities**  : Ayush Agrawal*, **Raghav Arora***, A. Datta, S. Banerjee, B. Bhowmick, K. Jatavallabhula, M. Sridharan, and M. Krishna
In: 32nd IEEE International Conference on Robot and Human Interactive Communication 2023() . Busan, Korea And,
In: Pretraining for Robotics Workshop at the 2023 International Conference on Robotics and Automation - ICRA ()

TEACHING AND LEADERSHIP EXPERIENCE

- **Teaching Assistantship**: CHE 253M: Measurement, Control, and Data Analysis Laboratory (UT Austin, 2025). Laboratory assistance, office hours, guidance, and scientific report grading.
- **Student Assistantship**: (PSD, BITS Pilani, 2020). Organized online industry exposure practice school for junior batches during COVID-19. This included creating a learning management system to allow companies and professors to create online sessions for students, create quizzes and grade them.
- **Volunteer Work**: Teaching underprivileged children at [Make a Difference](#) (2022-2023). 3 out of 5 students taught qualified university entrance exams and joined Bachelor of Computer Applications.
- **Volunteer Work**: Dog Walking and Playtimes at [Austin Pets Alive](#)
- **Projects' Head**: Led a team of 20+ developers for University wide applications and websites ([DVM](#), BITS Pilani, 2019-2020)

FELLOWSHIPS

- **Eastern European Machine Learning Summer School, 2023**: For attending the summer school and poster presentation on CLIPGraphs.
- **Pioneer Research Grant**: Research Fellowship for conducting my off-campus thesis work on 'Deep learning for quantum chemistry using Density Functional Tight-Binding Method' ([IPCD BITS Pilani](#), 2022)

LEARNING PROJECTS

- **Multi-touch detection using Tactile Sensing** [Dr. Lillian Chin](#)
Extended the capabilities of eFlesh, a 3D-printable, magnetic tactile sensor, to address the challenge of multi-contact localization and precise force estimation. We introduce a unified machine learning framework to distinguish single and multi-touch events data using magnetic data from embedded Hall-effect arrays.
- **Investigation of image mosaicing techniques for UAV navigation** [Dr. Meetha V. Shenoy](#)
Mosaicing aerial images captured by swarm of UAVs(Unmanned Aerial Vehicles), using feature detection and blending algorithms. We use Convolutional Neural Networks for extracting relevant features from aerial images, and match the common features to blend different images. RANSAC variation of BaySac algorithm is used for seamlessly blending the images, resulting in a large field of view for navigation.